

Referee Report, August 2011

This paper measures the impact of maternity leave on children’s outcomes at ages 4-5, using variation induced by Canada’s expansion of job-protected, partially compensated maternity leave. The authors identify the impact by comparing outcomes of children born in the four years after the 2001 policy change to children born in the four years before. Figure 1 and Table 3 show very clearly that the policy increased mother’s time at home during a child’s first year of life by just over two months. Tables 3 and 4 show that the policy had little effect on inputs into child development following the a child’s first birthday, although there are robust effects on full-time work. The authors use this evidence to motivate their instrumental variables approach—using exposure to the policy as an instrument for a mother’s time at home during a child’s first year of life.

In contrast to the OLS-based estimates, the IV estimates (Table 6) suggest that maternal care *hinders* performance on the Peabody Picture Vocabulary Test and the “Who Am I?” test. The significance of the estimated effect on PPVT is not robust to alternative controls for time, but the estimated effect on the Who Am I test is quite robust, indicating an effect of 0.19 standard deviations across the various specifications. There is no evidence of an effect on the “Know Your Numbers” test or on measures of aggression, anxiety, or hyperactivity, which are based on parent reports. For these “zero results,” the authors highlight that the standard errors are such that they could detect changes of 8-9 percent of a standard deviation at the five-percent level.

The motivation for the paper is quite convincing. Most of all, it is important that we consider the effect of parental care for a broader group of women than the typical study which uses variation based on policies that target the poor. And, from a developmental standpoint, it is informative to consider policies that affect parental care during specific windows, as this paper is able to do with its variation that affects the 6-12 month window of a child’s life. This paper certainly fills an important void in the literature.

My main concerns about this paper involve the validity of the empirical approach.

The identification strategy extremely simple (a positive attribute in many circumstances), estimating the pre-post difference in outcomes controlling for time with various functional forms. Given the simplicity of the research design, the authors could be much more transparent. We should be able to see all of the effects (or non-effects) in graphs that plot unadjusted means. While the authors go part of the way in this direction in figures 1 and 2, showing year of birth fixed effects, they could go much further in order to be more convincing. They would do well by be adopting many of the elements typically seen in regression discontinuity designs.

The authors have a paragraph on p. 18 saying that they find very similar results using an RD design but that they “prefer [their] instrumental variables approach because the number of observations close to the policy-change threshold in these survey data are not

sufficient to realize the advantages of a regression discontinuity design.” While the authors may not want to go in this direction altogether, I don’t find this argument compelling. First, instrumental variable approaches are regularly used in conjunction with RD designs via the “fuzzy-RD approach.” Second, not having a lot of data near the policy-change threshold is equivalent to a power issue that can be addressed in an RD-design by: (1) expanding the bandwidth or (2) taking a global approach. In fact, the estimates that the authors present in the paper can be thought of as global-RD estimates.

Regardless of how far in this direction the authors go, this paper would benefit a great deal from the types of specification checks that are common to RD papers. Does there appear to be manipulation of the running variable, i.e., are more children born following the policy change than before? (The authors speak to this question in a very indirect way in Table 3, estimating the effect on the number of siblings that children in the sample have.) Is there non-random sorting across the threshold, i.e., does the composition of children change following the policy change? These are first-order questions. All of the effects the authors find could be driven by non-random impacts on fertility. In particular, if the policy induced less able/healthy families to have children, that would explain the negative impacts of the policy expansion on children’s outcomes.

Other Comments

1. How big are the estimated effects and the effects that can be ruled out? It would be extremely useful if there were some frame of reference for the standard-deviation estimates that are cited. Because there is no reference, I don’t know how persuaded I am by the authors power to detect “small effects” where they find zero. It sounds good that the authors could detect an effect of 8-9 percent of a standard deviation at the five-percent level, but do we really anticipate that the policy could conceivably have an affect on behavioral outcomes that large? What effects have been found elsewhere on these outcomes?
2. One interesting aspect of the data, in my opinion, is that the pre-2001 maternity leave policy binding for a large share of women, but relatively few women exhausted the 12-month leave that was afforded to them by the policy change: “among women affected by the reform, the timing of the return to work changed from just under 6 months to almost 9 months post birth.” This has important implications for external validity. For example, we would expect different effects if the leave was expanded further or was slightly reduced.
3. Table 1 could be moved to the appendix.